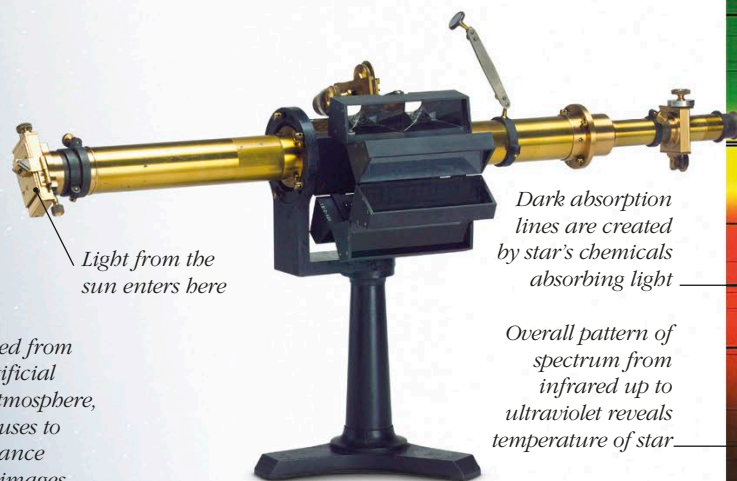


SPLITTING LIGHT

The spectroscope was invented in 1814 by the German physicist Joseph von Fraunhofer. It spreads light rays from stars onto different paths depending on their color or energy. Fraunhofer discovered dark, narrow lines in the sun's spectrum, which later scientists linked to light being absorbed by particular chemicals. Astronomers today use spectroscopy to study what celestial objects are made of.

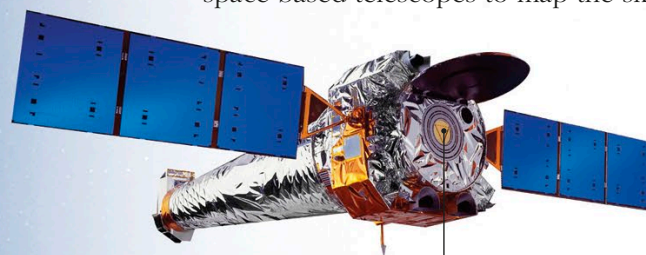


Solar spectroscope, 1881

Spectrograph of a star

SPACE TELESCOPES

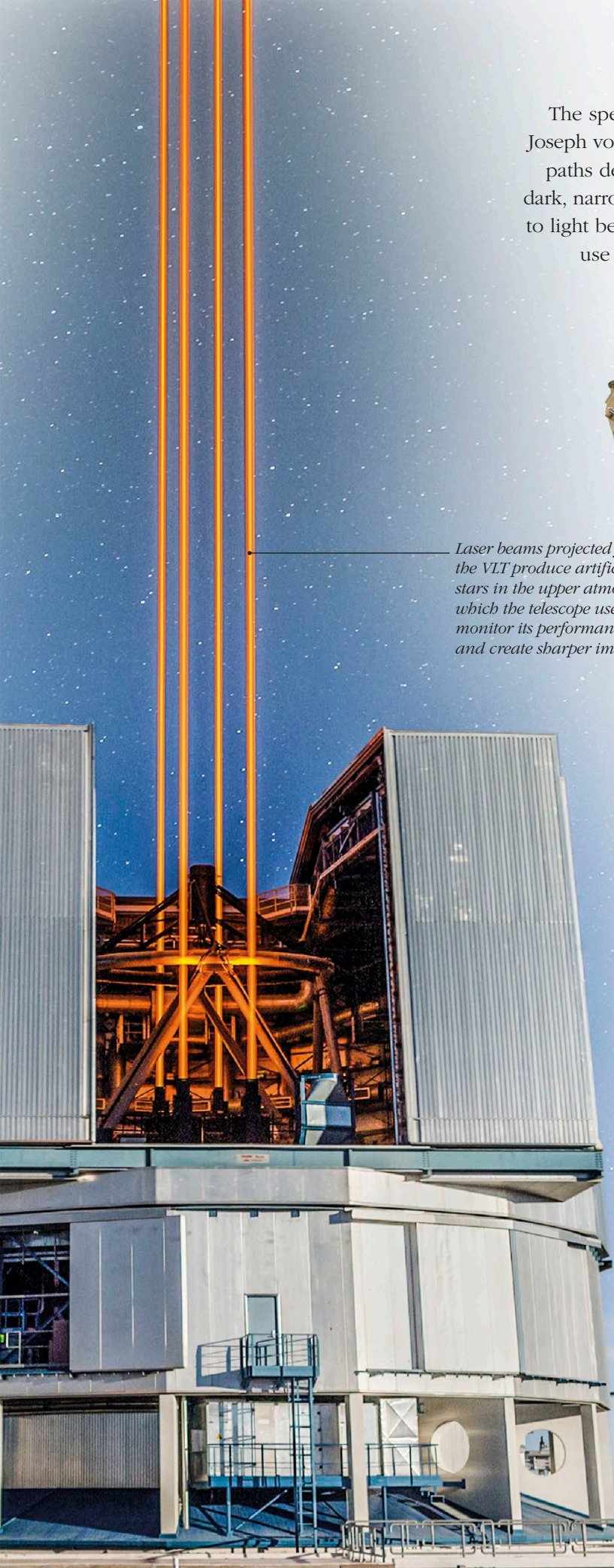
Visible light is just one part of the electromagnetic radiation emitted by cosmic objects. Things too cool to shine visibly still emit infrared radiation (heat), while very hot objects and violent events release high-energy ultraviolet, X-rays, and gamma rays. Earth's atmosphere blocks most of this radiation, so astronomers use space-based telescopes to map the sky.



Chandra X-ray Observatory

SMART TELESCOPES

All modern large telescopes are reflectors. Some have huge one-piece mirrors made of thin, lightweight materials, but most use interlocking honeycomb grids of smaller mirrors to reduce their weight. Computer-controlled motors behind the mirrors can adjust their overall shape to correct distortions and ensure perfect focus for whichever direction the telescope is pointing.



Laser beams projected from the VLT produce artificial stars in the upper atmosphere, which the telescope uses to monitor its performance and create sharper images.